

Interface Band Offsets as the Common Limit Across Four Lead-Free Perovskite Solar-Cell Absorbers: A Coupled Opto-Electrical COMSOL Study

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Abstract—Lead-free absorbers are the leading route to non-toxic halide-perovskite photovoltaics, but their device-simulation literature is dominated by one-dimensional, single-material optimizations that prescribe the optical generation profile, fix poorly known parameters, and rarely disclose parameter provenance, yielding projected efficiencies difficult to reconcile with experiment. We report a unified opto-electro-thermal finite-element (COMSOL) study of four chemically distinct lead-free absorbers (BaZrS₃, FASnI₃, Cs₂AgBiBr₆, Cu₂AgBiI₆) in a single fixed architecture, coupling full-wave electromagnetic optics to drift-diffusion transport, with every fixed parameter literature-sourced and provenance-graded and every uncertain parameter swept. Maxwell's equations are solved coherently on the real multilayer stack rather than a Beer-Lambert generation profile being prescribed, as is standard in this literature. Every result closes its AM1.5G-weighted optical energy budget to $\pm 0.00\%$ (worst per-wavelength residual 1.4%) and reproduces its photocurrent through two independent integration routes to within 0.05%, and the electromagnetic solver is verified against an independent transfer-matrix calculation to better than 0.1% absorbance at every wavelength, a check that also exposed and corrected a back-contact truncation artifact in one absorber. Under defect-free interfaces no device is limited by optical collection: the best collects 99.6% of its optical ceiling. Each is instead limited in voltage and fill factor by a band-offset cliff at a transport interface. The four materials partition into two mechanistic classes, electron-transport-layer conduction-band cliffs (FASnI₃, Cu₂AgBiI₆) and hole-transport-layer valence-band cliffs (BaZrS₃, Cs₂AgBiBr₆). Parasitic absorption in the transparent conductor is a first-order loss throughout, exceeding the absorber's own photocurrent in the most weakly absorbing case. Transport-layer band engineering, not further absorber optimization, is the operative design problem for this class.

Index Terms—Lead-free perovskite, photovoltaic cells, finite-element methods, numerical simulation, band offset, opto-electro-thermal modeling.

I. INTRODUCTION

EVERY halide perovskite solar cell that has crossed 26% certified power-conversion efficiency (PCE) is built on an absorber whose defining component, divalent lead, is water-soluble and toxic, an unresolved liability for the terawatt-scale, environmentally exposed deployment the technology is meant to serve [1]. Regulatory pressure and life-cycle concerns have made the replacement of lead the central materials problem of the field, and a chemically diverse set of lead-free absorbers has emerged in response [2]. These candidates are not variations on a theme: they span fundamentally different bonding, dimensionality, defect chemistry, and, crucially for

device design, band-edge energetics. That diversity is an opportunity, but it also means lessons from optimizing one lead-free absorber rarely transfer to another.

Four families frame the landscape. *Tin halide perovskites* such as FASnI₃ are the closest structural analogue to the lead compounds and the most advanced in single-junction devices [3], reaching 14.51% certified on 1 cm² fullerene-free architectures [4] and 16.65% through 2D/3D heterostructure engineering [5], though Sn²⁺ → Sn⁴⁺ oxidation remains their governing stability problem [6]. *Chalcogenide perovskites* such as BaZrS₃ replace halide with sulfide, a chemistry pursued for its thermal and moisture robustness, and their electronic structure is now characterized at the many-body level [7]. Their photovoltaic literature, however, remains overwhelmingly computational, with SCAPS-1D optimizations projecting 22 to 34% efficiencies [8], [9] that far outrun nascent experimental demonstrations. *Halide double perovskites* such as Cs₂AgBiBr₆ restore full inorganic stability, but wide indirect gaps and sluggish transport cap experimental single-junction PCE near 6% [10], [11] (a record that itself required gap-narrowing hydrogenation, while the same work's untreated control reached 0.55% [10]). Cs₂AgBiBr₆ is also the one absorber of the four with a prior three-dimensional optical-electrical simulation [12]. *Quaternary bismuth/silver iodides* such as Cu₂AgBiI₆ are strongly absorbing, air-stable, wide-gap semiconductors attractive for tandem top cells and indoor harvesting [13], [14], yet their experimental efficiencies remain near 2% [15], [16]. Across all four the pattern repeats: promising absorber physics, modest real devices, and a simulation literature whose optimism is hard to reconcile with experiment.

That reconciliation problem is largely methodological. The device-simulation studies populating this literature are predominantly one-dimensional, single-material optimizations that (i) prescribe the optical generation profile from an assumed absorption coefficient rather than solving Maxwell's equations for the real multilayer stack, (ii) fix critical and poorly known parameters at single borrowed values and then optimize around them, (iii) treat the cell as isothermal at 300 K, and (iv) rarely trace the origin of the inputs that determine the answer. Under these conditions it is unsurprising that an absorber capped near 6% in the laboratory can be projected above 25% on paper.

This work addresses those gaps with a unified opto-electro-thermal finite-element (COMSOL) study of the four absorbers

in a fixed architecture. Full-wave electromagnetic optics feed a finite-volume drift-diffusion transport model. Every fixed material parameter is literature-sourced with an explicit provenance status, and each genuinely uncertain quantity is treated as a sensitivity sweep. Three results follow. First, under defect-free interfaces no device is optically limited. Each is limited by a band-offset cliff at a transport interface, and the four partition into two mechanistic classes according to which contact carries that cliff. Second, parasitic absorption in the transparent conductor is a first-order loss in every device, exceeding the absorber's own photocurrent in the most weakly absorbing case. Third, methodologically, every optical result is validated three independent ways, including against an analytic transfer-matrix limit, a discipline that caught a geometric idealization which would otherwise have silently inflated one absorber's photocurrent by 4 %.

II. COMPUTATIONAL METHOD

All devices were simulated in COMSOL Multiphysics (v6.2 locally, v6.3 on the compute server) using a single self-consistent workflow coupling full-wave electromagnetic optics to finite-volume drift-diffusion transport, with an additional steady-state heat-transfer problem for $\text{Cs}_2\text{AgBiBr}_6$.

The governing principle for parameterization was to admit no unjustified point assumptions. Every fixed material parameter is literature-sourced and carries an explicit provenance status: **[EXP]** experimental, **[THEO]** computed with the level stated, **[SIM-LIT]** from published device simulations, **[DERIVED]** computed here from literature inputs through a stated parameter-free relation. Quantities that are genuinely unknown or that conflict across the literature (electron affinity χ , carrier lifetime τ , relative permittivity ϵ_r , mobility μ , doping) are **[SWEPT]** sensitivity axes. Full status-coded tables are in the Supplementary Material.

A. Device architecture

A fixed stack is used throughout: glass / FTO (600 nm) / TiO_2 ETL (50 nm) / absorber (300 nm) / Spiro-OMeTAD HTL (200 nm) / Au. The absorber is the only layer that changes between devices, so all comparisons are on identical optical, electrical and numerical footing. The optical unit cell is $350 \times 350 \text{ nm}^2$ with Floquet periodic boundaries on all four lateral faces, a perfectly matched layer above, and a scattering boundary condition at the illuminated plane. Fig. 1 shows the four absorbers' band edges against these fixed contacts, with the offset that limits each device highlighted.

B. Optical model

The electromagnetic problem is solved in the frequency domain, total-field formulation, for 61 wavelengths spanning 300 to 900 nm at 10 nm resolution under AM1.5G illumination [17]. Absorbed power density is integrated per layer, giving a closed optical budget: absorber plus each parasitic layer plus reflectance equals incident. Absorber optical constants are measured data for three of the four: tabulated ellipsometric ϵ for FASnI_3 [18], tabulated ellipsometry with a photothermal-deflection sub-gap tail for $\text{Cu}_2\text{AgBiI}_6$ [13], [19], and the

measured ellipsometric dielectric function for BaZrS_3 [20]. For $\text{Cs}_2\text{AgBiBr}_6$ no measured dielectric function of comparable spectral completeness has been published, and the HSE06-computed response of [21] is used instead, via its conductivity panel. That source's absorption-coefficient panel is internally inconsistent with its own conductivity and dielectric panels by a factor of 2π , which we confirmed by an independent re-digitization (the conductivity route gives $\alpha_{\text{true}}/\alpha_{\text{panel}} = 5.98$ to 6.21 over 3.8 to 4.4 eV, consistent with $2\pi = 6.28$), so the mutually consistent conductivity/dielectric panels are used and the α panel is not. The contacts use measured $\text{SnO}_2\text{:F}$ dispersion [22] and Au [23]. TiO_2 uses a Sellmeier index [24]. The four absorbers' optical constants are collected in Fig. 2.

Two aspects require care. Mishandling either silently corrupts the per-layer absorption partition. First, *Floquet periodic boundaries require the paired faces to carry identical surface meshes*. A free-tetrahedral mesher does not produce this, and the solver then interpolates across the mismatch without raising an error. The computed absorptance wanders with every remesh. On this stack, absorber absorptance at 700 nm took values 0.719, 0.663 and 0.463 on meshes differing by 2.4 % in element count, a 36 % swing. The fix is to mesh one face of each pair with a free-triangular operation and copy it onto the opposite face before the tetrahedral fill. The same comparison then gives 0.783103 versus 0.782888 (0.03 %). Critically, *the energy-conservation residual does not detect this error*: it compares total absorbed power against total incident flux and is blind to how that total is partitioned between layers, which is exactly what the photocurrent depends on.

Second, meshes are banded by wavelength (300 to 390, 400 to 590, 600 to 900 nm) because element count scales as λ^{-3} . Element size is set per material as $\lambda/(n_{\text{eff}} \text{ ppw})$ with $n_{\text{eff}} = |\tilde{n}|$, subject to a skin-depth clamp and a floor of three elements per layer thickness. Wavelengths at the band boundaries are solved on both neighbouring meshes as a seam check.

The AM1.5G weighting integrates the irradiance over each wavelength bin rather than sampling at bin centres, with outer bin edges clamped to the grid endpoints.

For the laterally uniform stacks considered here at normal incidence, the periodic three-dimensional problem is mathematically equivalent to a one-dimensional coherent multilayer problem. The finite-element framework is used because it extends unchanged to non-planar geometry. Section II-E verifies it against that analytic limit, which is also what exposed the one place where a geometric idealization mattered.

C. Electrical and thermal models

Transport is finite-volume drift-diffusion with Fermi-Dirac statistics, Shockley-Read-Hall recombination, and a radiative term $R = B(np - n_i^2)$ whose coefficient B is derived from the van Roosbroeck-Shockley relation [25] using the same optical constants that drive the optical model, making it parameter-free rather than fitted. Heterojunctions are treated in both quasi-Fermi-continuity and thermionic-emission modes. The two agree to 0.2 %. The broadband generation profile $G(z)$ from the optical stage is laterally averaged and supplied to the transport model. For $\text{Cs}_2\text{AgBiBr}_6$ a steady-state heat-transfer problem is coupled to the optical and electrical

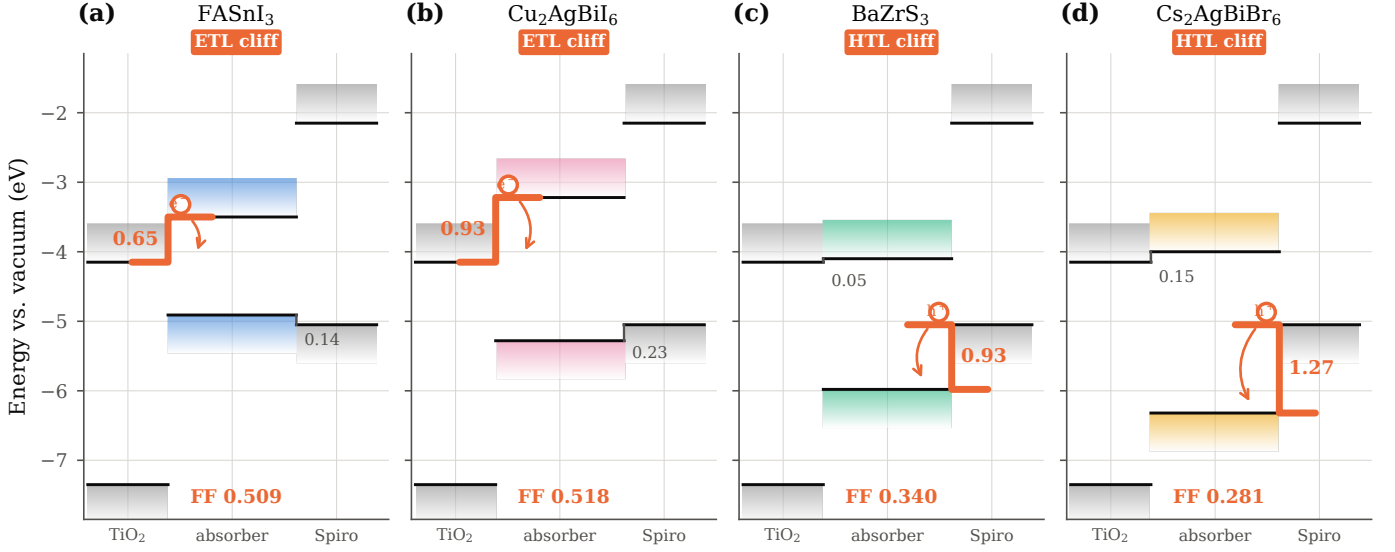


Fig. 1. Band alignment of the four devices against the vacuum level, with the conduction- and valence-band offsets at the two transport interfaces. The offset that limits each device is drawn as a literal cliff, with the carrier that falls over it and the fill factor it costs. The four partition into two classes: electron-contact cliffs (a, b) and hole-contact cliffs (c, d).

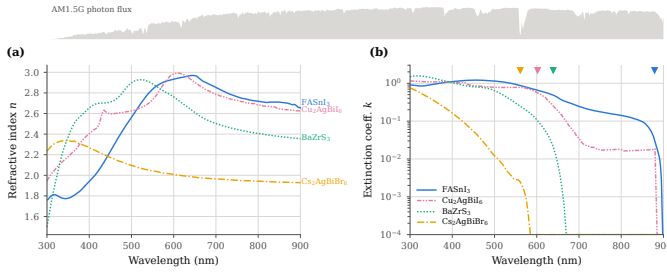


Fig. 2. Optical constants of the four absorbers over 300 to 900 nm: (a) refractive index n and (b) extinction coefficient k (logarithmic). The top strip shows the normalized AM1.5G photon flux on the shared wavelength axis, and the triangles in (b) mark each absorber's gap wavelength, where its collection window closes.

stages, following coupled opto-electro-thermal treatments of lead-based devices [26]–[28] and motivated by the reported sensitivity of self-heating to active-material properties [29], with per-layer thermal conductivities literature-cited [30]–[34]. No direct Spiro-OMeTAD measurement exists, so its value is a disclosed proxy from a structurally similar small-molecule organic semiconductor [35]. The thermal stage was closed before the final refinement of the optical inputs. The quoted temperature coefficient is a relative quantity, robust to the subsequent generation-profile rescale, and absolute operating temperatures are therefore not quoted.

D. Numerical procedure and validation

Every result carries three independent checks. The optical budget must close, and the depth integral of $G(z)$ must reproduce the photocurrent obtained independently from the spectral integral of layer absorptance. These are different routes through the same solve. Across all four absorbers the budget closes to $\pm 0.00\%$, the worst per-wavelength energy residual is -1.43% , the two photocurrent routes agree within

0.05%, and band seams agree to 0.020 to 0.096%. Third, an independent scalar transfer-matrix calculation with identical optical inputs and identical geometry reproduces the finite-element absorber absorptance to within 7×10^{-4} at every wavelength and the photocurrent to $\leq 0.02\%$, for all four absorbers (Fig. S7): a different solver checked against the analytic planar limit, not another route through the same solve. This third check exposed a real artifact: the optical model truncates the Au contact at 45 nm over a perfect electric conductor, justified by Au's 12 to 17 nm near-infrared skin depth, but Au's damping dips in its 450 to 570 nm interband window, and $\text{Cs}_2\text{AgBiBr}_6$ is the one absorber transparent exactly there. The truncation inflated its ceiling by 4.3%. All $\text{Cs}_2\text{AgBiBr}_6$ results below use the corrected thick-contact limit, computed with the validated transfer-matrix method and re-solved through transport (Supplementary Section S3).

Device J - V curves use continuation in applied bias. For $\text{Cu}_2\text{AgBiI}_6$ a 25 mV grid lands on spurious negative-differential-resistance branches. A 1 mV grid removes them, and only the dense result is quoted.

III. RESULTS

A. Full-wave optics and the current ceilings

Fig. 2 shows the optical constants and Fig. 3 the resulting absorptance, reflectance and closed loss budget. The optical current ceiling, the photocurrent at 100% collection, is 20.450, 15.083, 10.306 and 2.440 mA cm^{-2} for FASnI_3 , $\text{Cu}_2\text{AgBiI}_6$, BaZrS_3 and $\text{Cs}_2\text{AgBiBr}_6$ respectively (Table I).

BaZrS_3 's comparatively low ceiling follows from a deliberate data choice: the measured ellipsometric dielectric function [20] is adopted over Kramers-Kronig reconstructions of computed spectra, which zero-pad inside the material's own absorption range (Fig. S1). The measured data has weak near-edge absorption: the source states $E_g = 1.94$ eV, and the parameterized spectrum's absorptance falls below the 1%

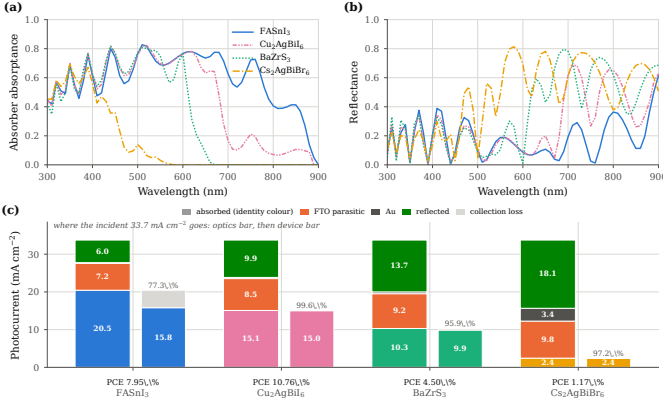


Fig. 3. Full-wave optical response. (a) Absorber absorptance and (b) stack reflectance versus wavelength. (c) Where the incident 33.74 mA cm^{-2} goes. For each absorber the optics bar partitions the incident current into absorbed, FTO, Au and reflected channels (closing to $\pm 0.00\%$ in every case), and the adjacent device bar carries the absorbed ceiling to the terminals as collected J_{sc} plus collection loss, with the resulting PCE beneath.

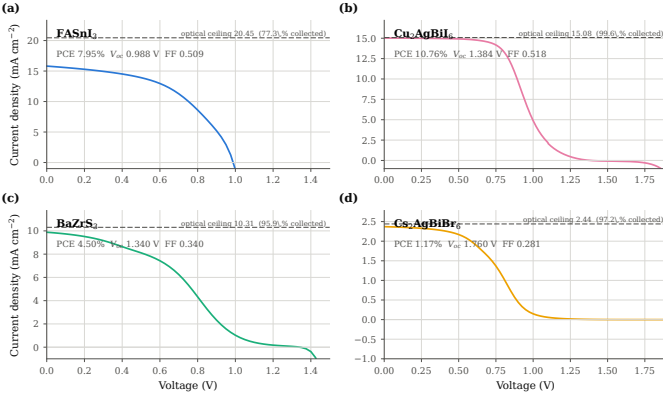


Fig. 4. Device J - V characteristics of the four absorbers on their full-wave generation profiles, with the extracted V_{oc} , FF and PCE annotated per panel. The dashed guide in each panel is that absorber's optical ceiling with the collected fraction, making the central claim directly visible: no device is limited by optical collection.

level beyond $\approx 660 \text{ nm}$ (1.88 eV), the onset visible in Fig. S6. The Cs₂AgBiBr₆ ceiling is quoted in the thick-contact limit, correcting the Au-truncation artifact exposed by the transfer-matrix check of Sec. II-E.

The loss budget (Fig. 3c) shows the parasitic cost of the transparent conductor is first-order for every absorber, ranging from 7.2 mA cm^{-2} (FASnI₃) to 9.8 mA cm^{-2} (Cs₂AgBiBr₆). In the weakly absorbing double perovskite the contact absorbs four times more photocurrent than the absorber does.

B. Devices of record

Fig. 4 shows the J - V characteristics and Table I the metrics, alongside each absorber's optical ceiling and the resulting collection efficiency.

No device is limited by optical collection. Cu₂AgBiI₆ collects 99.6% of its ceiling, Cs₂AgBiBr₆ 97.2%, BaZrS₃ 95.9%. Each is instead limited in voltage and fill factor.

The outlier, FASnI₃ at 77.3%, is instructive: it is the strongest absorber of the four, so its generation is concentrated

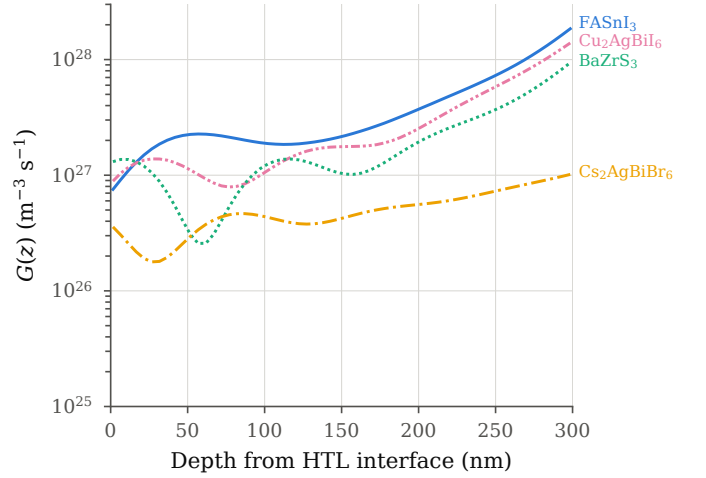


Fig. 5. Broadband AM1.5G generation profiles $G(z)$ through the 300 nm absorber, plotted from the hole-transport interface. The strong absorbers front-load generation toward the illuminated (electron- contact) face. Cs₂AgBiBr₆ generates almost uniformly.

toward the illuminated face (Figs. 5 and S6), far from the hole contact, and the difference is lost to bulk recombination in transit. Cs₂AgBiBr₆, absorbing weakly and generating almost uniformly through the depth, collects 97% despite far worse transport. Where in the depth carriers are deposited therefore matters as much as how many are generated, a distinction a prescribed exponential profile cannot express.

C. Two classes of interface cliff

The limiting mechanism in every device is a band-offset cliff at a transport interface, and the four materials partition into two classes (Fig. 1). FASnI₃ and Cu₂AgBiI₆ are limited by a *conduction-band cliff at the TiO₂ electron contact*, 0.65 eV and 0.93 eV respectively, which permits interfacial recombination and caps fill factor. BaZrS₃ and Cs₂AgBiBr₆ are limited by a *valence-band cliff at the Spiro-OMeTAD hole contact* [36], 0.93 eV and 1.27 eV, which blocks forward hole injection, produces an S-shaped J - V characteristic and collapses fill factor to 0.28 to 0.34, the hole-side analogue of the band-offset barrier mechanism established for conduction-band spikes at CIS window layers [37].

The partition is sharp. In each material the limiting offset exceeds the opposing one by at least a factor of three, and in BaZrS₃ and Cs₂AgBiBr₆ by more than eight, so the assignment does not depend on where the boundary between “cliff” and “spike” is drawn. It is also orthogonal to the chemistry: one tin halide and one bismuth iodide fall in the first class, one chalcogenide and one double perovskite in the second. Because electron affinity is a swept rather than fixed parameter here, the classification was checked across the swept range and is stable within it.

D. Two natures of the open-circuit voltage

An independent axis distinguishes how V_{oc} is set. For FASnI₃ the radiative term dominates and V_{oc} is pinned near its radiative limit. For the bismuth halides, bulk Shockley-Read-Hall recombination sets it.

TABLE I
OPTICAL CEILINGS AND DEVICE METRICS OF THE FOUR ABSORBERS

Absorber	Ceiling J_{ph}	J_{sc} (mA cm ⁻²)	V_{oc} (V)	FF	PCE (%)	Collection
FASnI ₃	20.450	15.811	0.988	0.509	7.95	77.3 %
Cu ₂ AgBiI ₆	15.083	15.017	1.384	0.518	10.76	99.6 %
BaZrS ₃	10.306	9.883	1.340	0.340	4.50	95.9 %
Cs ₂ AgBiBr ₆	2.440 [†]	2.371	1.760	0.281	1.17	97.2 %

Collection = $J_{sc}/\text{Ceiling}$, both in mA cm⁻². Cu₂AgBiI₆ on a uniform 1 mV continuation grid (Sec. III-E). [†]Thick-contact limit (Sec. II-E).

BaZrS₃ belongs to neither category. Its radiative coefficient, derived from the van Roosbroeck-Shockley relation over the measured optical constants (Fig. S5), is $B_{rad} = 5.37 \times 10^{-12} \text{ cm}^3 \text{ s}^{-1}$, and the device is insensitive to it: re-solving with a coefficient 435 $\times$ larger moves V_{oc} by only 3 mV and PCE by 0.02 points. BaZrS₃'s V_{oc} is therefore not radiatively limited. It is set by dark current through the Spiro injection barrier.

E. Numerical fidelity and disclosures

Three caveats must be stated explicitly.

Cu₂AgBiI₆ sub-gap absorption. 38.3% of this absorber's photocurrent (5.77 mA cm⁻²) originates below its 2.06 eV Tauc gap. This is not an artifact of the electromagnetic solve: the optical file is constructed from tabulated ellipsometry above 2.0 eV [19] and a photothermal-deflection tail shape below it [13], zeroed under 1.40 eV. Consequently the ratio of photocurrent to the detailed-balance ceiling at 2.06 eV reads 114%, which is meaningless. We therefore report the above-gap/sub-gap split rather than a ceiling fraction. Within that sub-gap current, 0.908 mA cm⁻² (6.0% of the total) arises from a segment where the absorption coefficient stops decaying and sits flat at $\approx 2.7 \times 10^3 \text{ cm}^{-1}$ for 130 nm before stepping to zero, a construction artifact rather than a physical Urbach tail. An optical ceiling credits all of these carriers at 100% collection, which is optimistic for tail states. Re-solving the device on restricted generation profiles bounds the assumption (Supplementary Section S7): with the artifact removed but the measured tail kept, PCE is 10.10%, and with no sub-gap collection at all it is 6.98%. The full-collection 10.76% of Table I is therefore the optimistic end of a [6.98, 10.76] % bracket whose defensible central value is 10.10%.

Cu₂AgBiI₆ open-circuit voltage is not grid-converged. The same model on a mixed 1 to 25 mV continuation grid gives $V_{oc} = 1.515 \text{ V}$ and on a uniform 1 mV grid 1.384 V, a 131 mV spread. Maximum power, V_{mpp} and J_{mpp} are identical to five significant figures (10.761 mW cm⁻², 0.786 V, 13.691 mA cm⁻²), so PCE is unaffected (Fig. S4). The cause is physical: the ETL cliff makes the pre- V_{oc} tail nearly flat, so a 0.08 mA cm⁻² numerical difference moves the zero crossing enormously. V_{oc} is poorly conditioned for this absorber. P_{max} -derived metrics are robust. A related gap-choice test (Supplementary Section S2) re-solved Cs₂AgBiBr₆ at its 2.21 eV optical onset and BaZrS₃ at the 1.94 eV gap stated by its optical source: PCE is invariant to four decimals in both, with the shift appearing entirely in V_{oc} ($\approx \Delta E_g$) and compensated

by the fill factor. The disclosed gap inconsistencies do not affect any efficiency conclusion.

Lossless transport layers. TiO₂ [24] and Spiro-OMeTAD are modeled with real-only refractive indices ($k = 0$), so their budget entries are exactly zero and all front parasitic absorption is attributed to the FTO. TiO₂ genuinely absorbs below $\approx 390 \text{ nm}$, so UV generation in the absorber is slightly overestimated. The treatment is identical across all four devices, so comparisons are unaffected.

IV. DISCUSSION

Three design consequences follow.

Transport-layer band engineering is the operative problem. With defect-free interfaces three of the four devices already collect 96 to 99.6% of their available photocurrent, and the fourth (FASnI₃, at 77%) loses the remainder to bulk recombination rather than optics. Further absorber optimization cannot address what limits any of them. The cliff taxonomy identifies which contact to change for which material: FASnI₃ and Cu₂AgBiI₆ need a shallower-affinity electron contact, BaZrS₃ and Cs₂AgBiBr₆ a deeper-HOMO hole contact. For BaZrS₃ neither Spiro-OMeTAD [38] nor CuO [39] provides both a deep HOMO ($\approx 5.9 \text{ eV}$) and a high conduction-band minimum, which is why that material's HTL selection remains unresolved.

Optical improvements must be evaluated in depth, not in aggregate. The FASnI₃/Cs₂AgBiBr₆ contrast shows that identical fractional gains in absorbed photons produce different device gains depending on where carriers are generated. Any light-management claim benchmarked only on absorptance may overstate its device benefit.

Parasitic absorption in the transparent conductor is first-order. It costs 7.2 to 9.8 mA cm⁻² across the four devices, and for Cs₂AgBiBr₆ the contact absorbs four times more photocurrent than the absorber does. Any optical improvement to these stacks must be assessed against this loss channel, not against reflection alone.

For Cs₂AgBiBr₆ the closed thermal loop indicates a benign relative temperature coefficient of $\approx -0.12 \% \text{ K}^{-1}$. This figure was obtained with the pre-refinement optical inputs (Sec. II) and is quoted here, rather than among the results, as a relative quantity robust to the generation rescale.

Limitations. Interfaces are defect-free, so reported efficiencies are upper bounds. No literature interface-recombination data exists for these material pairs, and junction recombination is not representable by the solver's surface-recombination feature in this formulation (Supplementary Section S8). Because the limiting mechanism is carrier accumulation at a band-offset

cliff, the upper-bound label is load-bearing. Transport layers are optically lossless. The thermal loop is closed only for $\text{Cs}_2\text{AgBiBr}_6$. $\text{Cu}_2\text{AgBiI}_6$'s V_{oc} carries the convergence caveat above, and its optical constants include a sub-gap tail whose device consequences we do not model as distinct from band-edge carriers.

V. CONCLUSION

A unified opto-electro-thermal study of four chemically distinct lead-free absorbers in a fixed architecture, with coherent full-wave optics verified against the transfer-matrix limit, shows that none is limited by optical collection under defect-free interfaces. Each is limited by a band-offset cliff at a transport interface, and the four partition into electron-contact and hole-contact cliff classes. Every optical result closes its energy budget exactly, agrees between two independent photocurrent routes, and is verified against an analytic transfer-matrix limit, a discipline that exposed and corrected a geometric idealization which would otherwise have silently inflated one absorber's photocurrent by 4%. Parasitic absorption in the transparent conductor is a first-order loss in every device. Transport-layer band engineering, not further absorber optimization, is the operative next step for this class of materials.

DATA AVAILABILITY

All simulation inputs, build scripts, result files and figure-generation code are available at <https://github.com/mohiuddinsciencelover-glitch/leadfree-pv-3d-simulation>, together with `REPRODUCE.md`, which documents the full pipeline end to end.

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